

Omar Ahmad Elhakim

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Education

Ain Shams University

BSc in Computer Science, Scientific Computing Department | GPA: 3.3

Graduated June 2026

Experience

Zinad

July 2024 – Present

AI Engineer (Part-Time)

- Built a phishing simulation engine that generates themed emails and working fake landing pages from the parameters a user picks, driving **Ollama** from a **Laravel** backend and rendering the emails with **MJML**.
- Wrote an email analyzer in **Python**, **LangChain** and **Ollama** that flags phishing from tone, context and grammatical anomalies; its verdicts feed the platform's security metrics.
- Developed an internal **RAG** system that answers employee questions from the company's policy PDFs, splitting the documents with **recursive character chunking** with overlap, embedding the chunks into a **FAISS** vector store and retrieving the passages relevant to each question.
- Shipped a chatbot into the live application with **LangChain** and **Ollama**, streaming replies token by token instead of making the user wait for the full answer.

Projects

Neuromarketing: EEG Emotion Recognition [↗](#) — Graduation Project

- Classified emotion from raw **EEG** brain signals on the **SEED-IV** dataset, tuning nine deep-learning architectures (**DaViT**, **DGCNN**, **CCNN**, **BFENet**, **TSCeption** and others) in **PyTorch** under subject-dependent and cross-subject protocols, reaching **96%** accuracy on binary and **85%** on 4-class emotion recognition.
- Combined the dataset's precomputed **differential entropy** features with our own: per-band differential entropy extracted through **db4 wavelet** decomposition, and hemispheric asymmetry (**DASM**) features derived from it.
- Built the raw-signal preprocessing pipeline in **SciPy** and **NumPy** (Butterworth bandpass, downsampling, z-score normalization, windowing), and re-ran the experiments on the **14 channels** of a consumer Emotiv EPOC+ headset to test viability on affordable hardware.

Blood Cell Detection with YOLO [↗](#)

- Located and classified red cells, white cells and platelets in blood-smear microscope images, fine-tuning pretrained **YOLOv8** and **YOLOv5** detectors on 364 annotated slides and benchmarking the two generations on the same split at **0.93 mAP@50**.
- Wrote the dataset pipeline behind the training run, converting **~5000** corner-coordinate bounding boxes into normalized **YOLO** center format with **OpenCV**.
- Evaluated per class, holding **0.98 mAP@50** on white cells despite being outnumbered ten to one by red cells.

RAG [↗](#)

- Built a document Q&A service in **FastAPI** and **LangServe**, splitting a company policy corpus into overlapping chunks with a **recursive character splitter**, embedding them with **word2vec**, indexing them in a **Chroma** vector store, and grounding **Gemini** on the chunks retrieved for each question.
- Added a phishing screener that runs an email body through a self-hosted **Llama** on **Ollama**, returning nine checks (urgency language, requests for confidential information, character-encoding tricks such as Cyrillic lookalikes, and others) as structured **JSON** with English and Arabic justifications, covered by end-to-end API tests.
- Extended retrieval to databases with a **text-to-SQL** chain that feeds the live **MySQL** schema to the model, executes the generated query and phrases the result in plain English.

Meal Recognition and Calorie Estimation [↗](#)

- Estimated the calories in a meal photo in **PyTorch**, chaining five trained models behind one pipeline: an **EfficientNet** that sorts the photo into food or fruit, then a **ResNet** that names the fruit or a **Siamese** network that names the dish.
- Handled unseen dishes with **one-shot** recognition rather than a fixed label set, matching a query against reference embeddings by distance and returning no match when nothing is close enough, so new dishes can be added from a single example without retraining.
- Segmented the fruit from its background with a **U-Net** at **0.99 IoU**, and added a 31-class segmentation model in **TensorFlow** to label each fruit type pixel by pixel.

Technical Skills

Languages: Python, C/C++, C#, Java, PHP, JavaScript, SQL, Bash

Tools/Frameworks: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, SciPy, OpenCV, LangChain, FastAPI, Laravel, Docker, Git, Linux, Django

Training & Development

- MEC Model of Leadership:** Completed an extensive leadership program (160 hours) covering emotional intelligence, effective communication, self-awareness (50 hours), the Enneagram, problem-solving, and team building.